

3 Literature Review

Researchers have studied whether the tone of financial news can help predict stock returns for nearly two decades, and the methods have developed from simple word counts to context-aware deep learning models. Tetlock (2007) used the Harvard psychosocial dictionary to measure negative language in a daily Wall Street Journal column and found that high media pessimism was followed by downward price pressure, unusually high trading volume, and a later return toward fundamentals. Loughran and McDonald (2011) showed why general-purpose dictionaries can be unreliable in finance. They found that many words labeled as negative, including tax, cost, liability, and capital, are often neutral in financial documents. Their finance-specific word lists became a widely used baseline because they better matched the meaning of financial language. However, dictionary methods still treat text as an unordered collection of words, so they can miss context, negation, and the meaning of a complete sentence.

Later research used supervised learning, event representations, and social-media data to capture more complex relationships between language and market behavior. Ke, Kelly, and Xiu (2019) developed a supervised text-mining framework that identified return-predictive terms, weighted them through a topic model, and combined them into article-level sentiment scores. Using Dow Jones Newswires, they found that task-specific sentiment scores provided stronger predictive signals than dictionary measures and commercial sentiment products. Bollen, Mao, and Zeng (2011) analyzed mood in Twitter messages and reported that one mood dimension improved directional prediction of the Dow Jones Industrial Average, although later studies have found social-media results to be less stable. Ding, Zhang, Liu, and Duan (2015) represented financial news as structured events and used event embeddings with a convolutional neural network. Their model outperformed bag-of-words baselines when predicting both market-index and individual-stock movements.

Commercial systems also show that news sentiment is used outside academic research.

RavenPack has provided entity-tagged news analytics, sentiment measures, and event classifications since 2003 for applications such as quantitative signal generation and risk management. LSEG MarketPsych Analytics converts large volumes of news and social-media content into sentiment, emotion, and topic scores at minute, hourly, and daily frequencies, with historical data extending back to 1998. These systems demonstrate that news-derived sentiment has become an established input for financial analysis rather than remaining only a research concept.

Existing research also shows that the value of a sentiment signal depends on the prediction or holding horizon. Tetlock (2007) found that the price effects connected to media pessimism were temporary and generally reversed within about one trading week. Heston and Sinha (2017) examined this relationship more directly and reported that daily news predicted returns over the next one to two days, while news aggregated over a week predicted returns for as long as a quarter. Ke et al. (2019) similarly found that news-based strategies performed best at shorter rebalancing intervals and weakened as the interval increased from weekly to monthly and quarterly. Together, these findings show that sentiment signals decay over time and that their usefulness can change depending on how the news is aggregated and how long a position is held.

Transformer-based sentiment models developed from earlier advances in natural language processing. Vaswani et al. (2017) introduced the transformer architecture, which uses self-attention to represent each word in relation to the other words in the same context. Devlin, Chang, Lee, and Toutanova (2019) extended this approach through BERT, a model pre-trained on large unlabeled text collections and then fine-tuned on smaller labeled datasets. BERT produced state-of-the-art results across several language-understanding tasks and reduced the amount of labeled data needed for specialized applications. Araci (2019) adapted BERT to financial language by further pre-training it on a financial news corpus and fine-tuning it on the Financial PhraseBank created by Malo et al.

(2014). FinBERT improved sentiment-classification accuracy over earlier methods, especially for context-dependent financial statements that dictionary approaches often classify incorrectly. Similar transformer-based systems have also been used commercially. For example, LSEG has applied a RoBERTa-based architecture to score sentiment and emotion in company transcripts at the sentence level across thousands of companies.

Sequence models and nonlinear machine learning methods have also been applied to financial prediction. Hochreiter and Schmidhuber (1997) developed long short-term memory networks to help recurrent models retain information across longer sequences. Fischer and Krauss (2018) later applied an LSTM to daily returns for S&P 500 companies and reported stronger directional-prediction performance than random forests, a deep feedforward network, and logistic regression. Gu, Kelly, and Xiu (2020) compared multiple machine learning methods for estimating equity risk premiums and found that neural networks and regression trees produced the largest improvements over linear models because they could capture nonlinear relationships and interactions among predictors. This research provides examples of both transformer-based financial sentiment analysis and sequence-based market prediction, showing that the two methods are well suited for systems that combine text, sentiment, and time-dependent market data.

Research on financial machine learning also emphasizes that model evaluation must account for the risks of backtest overfitting and biased historical testing. Sharpe (1994) formalized the Sharpe ratio as a standard measure of risk-adjusted performance. Bailey, Borwein, Lopez de Prado, and Zhu (2014) showed that testing many strategy variations on the same historical data can produce apparently profitable results by chance. Lopez de Prado (2018) described additional sources of inflated performance, including look-ahead bias, survivorship bias, and omitted transaction costs, and recommended point-in-time data alignment and strict separation of tuning and evaluation samples. As a result, similar financial sentiment projects are

commonly evaluated not only by predictive accuracy or total return, but also by whether the backtest controls for data leakage, selection bias, trading costs, and risk.

Use of AI tools: I used Anthropic's Claude (Anthropic, 2026) to help locate and organize the literature and assist with an initial draft of this section. I verified each cited source and its reported findings and take full responsibility for the final content.

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